

Equivariant Systems Theory and Observer Design for Autonomous Systems

R. Mahony J. Trumpf T. Hamel



Australian
National
University



Module M17, EECI International Graduate School on Control
Zaragoza, 22 June – 26 June 2026

Outline

- 1 Inertial Navigation Systems
 - Inertial Frames and Galilean Transformations
 - Galilean Kinematics
 - Gravity
 - Isochronous frames
 - INS kinematic model
- 2 Equivariant Filter
 - Problem Setup
 - Computing the Lift
 - Error formulation
- 3 Computing Error Linearisations
 - State Linearisation
 - Output Linearisation
- 4 Perspective

Outline

- 1 Inertial Navigation Systems
 - Inertial Frames and Galilean Transformations
 - Galilean Kinematics
 - Gravity
 - Isochronous frames
 - INS kinematic model
- 2 Equivariant Filter
 - Problem Setup
 - Computing the Lift
 - Error formulation
- 3 Computing Error Linearisations
 - State Linearisation
 - Output Linearisation
- 4 Perspective

Inertial Navigation Systems

An **Inertial Navigation System** (INS) is an algorithm that takes measurements from

- **Inertial Measurement Unit** (IMU) that measures angular velocity and proper acceleration.
- **Global Navigation Satellite Systems** (GNSS) that measures world fixed coordinates.

and uses these measurements to estimate the

- **Navigation states** consisting of attitude, position and velocity, (and global time)
- **Bias states** consisting of biases in the IMU and other measurement devices (intrinsic calibration).
- **Calibration states** consisting of sensor locations and orientation in the body-frame (extrinsic calibration).

Inertial Measurement Unit

The starting point is to understand what is meant by an inertial frame.

An inertial frame is a non-accelerating, non-rotating, frame in the absence of gravity.

Any frame moving at a constant velocity relative to an inertial frame is also inertial and the concept is intrinsic, that is independent of which frame is used as reference.

Inertial reference frames provide a consistent 'stage' for describing motion—physical laws are *invariant* in the sense that they retain the same form regardless of an observer's inertial frame of reference.

Relative Galilean pose

Consider two frames $\{A\}$ and $\{B\}$ in space.

Then we can describe the pose of frame $\{B\}$ with respect to $\{A\}$ by a rotation and a position

$$R = {}^A R_B, \quad p = {}^A p_B$$

In addition, to understand inertial motion one adds relative velocity and time offset

$$v = {}^A v_B, \quad t = {}^A t_B$$

We will call ${}^A F_B$ a Galilean pose

$$F = {}^A F_B = (R, v, p, t) = ({}^A R_B, {}^A v_B, {}^A p_B, {}^A t_B)$$

Inertial frames

If frame $\{A\}$ and $\{B\}$ are inertial then

$$\dot{R} = 0$$

$$\dot{v} = 0$$

$$\dot{p} = v$$

$$\dot{t} = 1$$

That is $\{B\}$ can be translating at a constant speed but cannot be accelerating or rotating in $\{A\}$.

Principle of Absolute Time

Newton introduced a principle of absolute time. That is, time runs at the speed for any physical observer $\dot{t} = 1$.

A consequence of this principle is that the speed of light is infinite.

Galilean Transformation

A Galilean transformation is a transformation of Galilean space-time - an extension of rigid-body transformations.

Let (Q, b, q, δ) be a set of parameters

$Q \in \mathbf{SO}(3)$ a rotation

$b \in \mathbb{R}^3$ a velocity boost

$q \in \mathbb{R}^3$ a translation

$\delta \in \mathbb{R}$ a time shift

A Galilean transformation (Q, b, q, δ) of a Galilean frame (R, v, p, t) is $(Q, b, q, \delta) \cdot (R, v, p, t)$ given by

$$\begin{aligned} R &\mapsto QR, & v &\mapsto Qv + b \\ p &\mapsto Qp + q + bt, & t &\mapsto t + \delta \end{aligned}$$

Galilean Group

The set of all Galilean transformations (Q, b, q, δ) is a Lie-group **Gal**(3)

$$\begin{aligned} (Q_1, b_1, q_1, \delta_1)(Q_2, b_2, q_2, \delta_2) \\ &= (Q_1 Q_2, Q_1 b_2 + b_1, Q_1 q_2 + q_1 + \delta_2 b_1, \delta_1 + \delta_2) \\ (Q, b, q, \delta)^{-1} &= (Q^\top, -Q^\top b, -Q^\top q - \delta b, -\delta) \\ I &= (I_3, 0, 0, 0) \end{aligned}$$

The matrix realisation of the group is

$$\begin{pmatrix} Q & b & q \\ 0 & 1 & \delta \\ 0 & 0 & 1 \end{pmatrix}$$

Inertial Frames

If $\{A\}$ and $\{B\}$ are inertial frames then

- ① $(Q, b, q, \delta) \cdot (R, v, p, t)$ is also an inertial frame
- ② $(R, v, p, t) \in \mathbf{Gal}(3)$ are group coordinates of $\{B\}$ with respect to $\{A\}$. We will write the Galilean frame $\{B\}$ with respect to $\{A\}$ in homogeneous coordinates

$$F = \begin{pmatrix} R & v & p \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}$$

The group action of $\mathbf{Gal}(3)$ on Galilean frames is

$$\begin{pmatrix} Q & b & q \\ 0 & 1 & \delta \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} R & v & p \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} QR & Qv + b & Qp + q + bt \\ 0 & 1 & \delta + t \\ 0 & 0 & 1 \end{pmatrix}$$

Non inertial frames

The kinematics of a general frame is given by

$$\dot{R} = R\omega^\times$$

$$\dot{v} = Ra$$

$$\dot{p} = v$$

$$\dot{t} = 1$$

Note that setting $\omega = 0 = a$ and solving for (R, v, p, t) one recovers an inertial frame

$$F(t) = \begin{pmatrix} R(t) & v(t) & p(t) \\ 0 & 1 & t(t) \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} R_0 & v_0 & p_0 + tv_0 \\ 0 & 1 & t + t_0 \\ 0 & 0 & 1 \end{pmatrix}$$

The angular velocity ω and acceleration a characterise the ‘non-inertial’ change in the Galilean frame F .

Inertial Measurement Unit

An inertial measurement unit IMU measures the non-inertial motion of a Galilean frame to which it is attached

- inertial angular velocity $\omega \in \mathbb{R}^3$
- proper acceleration $a \in \mathbb{R}^3$ that encodes the rate of change of velocity.

In practice, any measurement device has limitations. The most important for IMU is bias and noise

$$\begin{aligned}\omega_{\text{meas}} &= \omega_{\text{true}} + b_{\omega} + \mu_{\omega}, & \mu_{\omega} &\in \mathbf{N}(0, M_{\omega}) \\ a_{\text{meas}} &= a_{\text{true}} + b_a + \mu_a, & \mu_a &\in \mathbf{N}(0, M_a)\end{aligned}$$

With bias dynamics

$$\begin{aligned}\dot{b}_{\omega} &= 0 \\ \dot{b}_a &= 0\end{aligned}$$

Gravity

Gravity is actually curvature of space time! The consequence of not modelling curved space time in Galilean space-time is that gravity does not fit nicely in our framework.

The acceleration necessary to model Galilean kinematics is

$$a = a^p + g$$

where

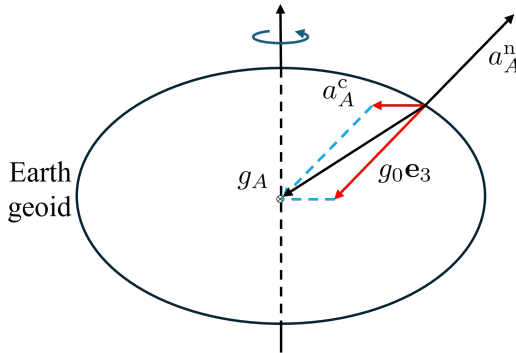
a^p proper acceleration

g gravitational acceleration

The inertial sensor only measures proper acceleration

$$a_{\text{meas}} = a^p + b_a + \mu_a, \quad \mu_a \sim \mathbf{N}(0, M_a)$$

Earth Geoid



The consequence of this means that understanding gravity on the surface of the earth is quite sophisticated.

In robotics, we work with apparent gravity $g = g_0 \mathbf{e}_3$ in most applications and model it in the Earth frame.

Isochronous Frames

In practice, we do not measure the pose of a frame at some time in the future, we measure the location of the frame at the present time. That is the reference frame $\{A\}$ is also moving in time with respect to a Galilean reference $\{0\}$

$${}^0F_A = \begin{pmatrix} I_3 & 0 & 0 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix} \quad {}^0F_B = \begin{pmatrix} R & v & p \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}$$

This the unique stationary inertial frame that is confounded with $\{0\}$ at time $t = 0$.

The frame $\{B\}$ is **isochronous** to $\{A\}$ if ${}^0t_B = {}^0t_A$. That is

$${}^AF_B = \begin{pmatrix} R & v & p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Extended Pose Group

Note that any two Galilean transformations

$$(Q_1, v_1, q_1, 0)(Q_2, v_2, q_2, 0) = (Q_1 Q_2, Q_1 v_2 + v_1, Q_1 p_2 + p_1, 0)$$

with $\delta_1 = 0 = \delta_2$ also have $\delta_3 = 0$. The inverse is

$$(Q, v, q, 0)^{-1} = (Q^\top, -Q^\top v, -Q^\top p, 0)$$

The set of all such transformations is termed the **extended pose** group or **isochronous Galilean** group.

$$\begin{aligned} \mathbf{SE}_2(3) &= \{(Q, b, q, 0) \mid Q \in \mathbf{SO}(3), b, q \in \mathbb{R}^3\} \\ &= \left\{ \begin{pmatrix} Q & b & q \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mid Q \in \mathbf{SO}(3), b, q \in \mathbb{R}^3 \right\} \end{aligned}$$

Extended Pose Action

The extended pose group acts on extended poses (isochronous Galilean frames) in the natural manner.

$$\begin{pmatrix} Q & b & q \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} R & v & p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} QR & Qv + b & Qp + q \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The extended pose group is the structure that is most commonly used to study inertial navigation after it was rediscovered by Bonnabel *et al.* in 2014 called the special Euclidean double isometry group.

INS Kinematics

We will use the notation $\omega = \omega_{\text{meas}}$ and $a = a_{\text{meas}}$.

The (noise free) INS kinematics are

$$\dot{R} = R(\omega - b_\omega)^\times \quad (1a)$$

$$\dot{v} = R(a - b_a) + g \quad (1b)$$

$$\dot{p} = v \quad (1c)$$

$$\dot{b}_a = 0 \quad (1d)$$

$$\dot{b}_\omega = 0 \quad (1e)$$

Navigation states are (R, v, p) .

Bias states are $b = (b_\omega, b_a)$.

The Inertial Measurement Unit measures (ω, a) .

Gravity $g = g_0 \mathbf{e}_3$ is assumed to be known.

Galilean symmetry

Putting this in matrix algebra one can compute $\frac{d}{dt}F = (\dot{R}, \dot{v}, \dot{p}, 0)$.

$$\begin{pmatrix} \dot{R} & \dot{v} & \dot{p} \\ 0 & \dot{1} & 0 \\ 0 & 0 & \dot{1} \end{pmatrix} = \begin{pmatrix} \dot{R} & \dot{v} & \dot{p} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ = \begin{pmatrix} R(\omega - b_\omega)^\times & R(a - b_a) + g & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and

$$\begin{pmatrix} \dot{b}_\omega^\times & \dot{b}_a & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Extended pose model

$$\begin{aligned}
 & \begin{pmatrix} R(\omega - b_\omega)^\times & R(a - b_a) + g & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} R & v & p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} (\omega - b_\omega)^\times & (a - b_a) & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
 &+ \begin{pmatrix} 0 & g & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} R & v & p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &+ \begin{pmatrix} R & v & p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} R & v & p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}
 \end{aligned}$$

Extended pose kinematics

Define

$$F = \begin{pmatrix} R & v & p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad G = \begin{pmatrix} 0 & g & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad N = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

$$U(\omega, a) = \begin{pmatrix} \omega^\times & a & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad B = U(b_\omega, b_a) = \begin{pmatrix} b_\omega^\times & b_a & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Then

$$\begin{aligned} \dot{F} &= FN - NF + GF + F(U(\omega, a) - B) \\ \dot{B} &= 0 \end{aligned}$$

for $F \in \mathcal{SE}_2(3)$ and $B \in \mathfrak{se}_2(3)$

Extended Pose Kinematics

Inertial Navigation System Kinematics

$$\begin{aligned}\dot{F} &= FN - NF + GF + F(U(\omega, a) - B) \\ \dot{B} &= 0\end{aligned}$$

for $F \in \mathcal{SE}_2(3)$ and $B \in \mathfrak{se}_2(3)$.

Note that the kinematics are posed on the torsor of $\mathcal{SE}_2(3)$. That is the same manifold structure as $\mathbf{SE}_2(3)$ but without the group structure.

The bias $B \in \mathfrak{se}_2(3)$ is on vector space of $\mathfrak{se}_2(3)$ rather than the Lie-algebra, however, we don't make a notational difference.

Rotating Earth INS Kinematics

If one needs to model the kinematics with respect to an isochronous rotating frame $\{A\}$ fixed to the surface of the Earth, then

$$\begin{aligned}\dot{F} = & FN - NF - U(\Omega_E, -g_0\mathbf{e}_3)F + F(U(\omega, a) - B) \\ & (U(0, g_B) - U(0, g_A))F\end{aligned}$$

where

- Ω_E is the Earth angular velocity,
- $g_0 \approx 9.8$ is the gravitational constant
- g_A is the gravitation field at the origin of the reference frame $\{A\}$ (expressed in reference coordinates).
- g_B is the gravitational field at the origin of the body frame $\{B\}$ (expressed in reference coordinates).

Outline

- 1 Inertial Navigation Systems
 - Inertial Frames and Galilean Transformations
 - Galilean Kinematics
 - Gravity
 - Isochronous frames
 - INS kinematic model
- 2 **Equivariant Filter**
 - Problem Setup
 - Computing the Lift
 - Error formulation
- 3 Computing Error Linearisations
 - State Linearisation
 - Output Linearisation
- 4 Perspective

Equivariant Filter

Recall the form for the Equivariant Filter

Predict:

$$\hat{X} \leftarrow \hat{X} \exp \left(\Lambda(\phi_{\xi}(\hat{X}), u) dt \right),$$

$$\Sigma \leftarrow \mathbf{A}_t \Sigma \mathbf{A}_t^{\top} + B_t \mathbf{M} B_t^{\top},$$

Update:

$$K := \left(\Sigma \mathbf{C}^{\top} (\mathbf{C} \Sigma \mathbf{C}^{\top} + \mathbf{N})^{-1} \right)$$

$$\Sigma \leftarrow (I - K \mathbf{C}) \Sigma,$$

$$\Delta := D_{E|I} \phi_{\xi}(E)^{\dagger} K \left(\rho(\hat{X}^{-1}, y) - \hat{y} \right),$$

$$\hat{X} \leftarrow \exp(\Delta) \hat{X},$$

Equivariant Filter

To implement this filter we will need to

- Pick an origin ξ°
- Define $\Lambda(\phi_\xi(\hat{X}), u)$ and define the observer lifted system.
- Compute the error system.
- Compute the linearisation A_t, B_t of the error system.
- Compute the linearisation C of the equivariant innovation.
- Define a pseudo inverse $D_{E|I} \phi_\xi^\dagger$.

The rest is matrix algebra.

Set up

We will first consider the bias free case $B = 0$

$$\dot{F} = FN - NF + U(0, g_0 \mathbf{e}_3)F + FU(\omega, a)$$

We will use $F \in \mathcal{SE}_2(3)$ is an extended pose to replace ξ as the state variable in the formula to keep the intuition of the state variable.

We use $U(0, g_0 \mathbf{e}_3) = G$ to emphasise that g_0 is treated as an input.

The inputs (ω, a) in the system model are noise free. When we build the model for the observer we will need to model noise in the inputs.

$$\begin{aligned} \omega_y &= \omega + \mu_\omega, & \mu_\omega &\sim \mathbf{N}(0, M_\omega) \\ a_y &= a + \mu_a, & \mu_a &\sim \mathbf{N}(0, M_a) \end{aligned}$$

Set up

The symmetry group is the extended pose group $\mathbf{SE}_2(3)$. We will denote elements of this group as $\hat{X} \in \mathbf{SE}_2(3)$ since these elements represent observer state.

The group action is right multiplication

$$\begin{aligned}\phi : \mathbf{SE}_2(3) \times \mathcal{SE}_2(3) &\rightarrow \mathcal{SE}_2(3) \\ \phi(\hat{X}, F) &:= F\hat{X}\end{aligned}$$

where

$$F = \begin{pmatrix} R & v & p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \hat{X} = \begin{pmatrix} \hat{Q} & \hat{b} & \hat{q} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Where $\hat{X} \in \mathbf{SE}_2(3)$ is the filter state and $F \in \mathcal{SE}_2(3)$ is the system state.

Set up

We will choose the origin $\mathring{F}$ to be

$$\mathring{F} = \begin{pmatrix} I_3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_5$$

Then the state estimate is

$$\hat{F} = \phi_{\mathring{F}}(\hat{X}) = \mathring{F}\hat{X} = \hat{X}$$

where $\mathring{F} = I_5$. This is a typical example of a free group action where the group and the system state are confounded.

Specifically, the stabiliser is trivial

$$\text{stab}_{\phi}(\mathring{F}) = \{I\}$$

Trivialised lift

The inputs $u = (\omega, a, g_0)$ are the angular velocity, acceleration, and gravitational constant.

trivialised lift

$$\Lambda(F, u) = \text{Ad}_{F^{-1}}(U(0, g_0 \mathbf{e}_3) - N) + U(\omega, a) + N$$

noting that $G = U(0, g_0 \mathbf{e}_3)$.

$$\begin{aligned} D\phi_F[\Lambda(F, u)] &= D_{\hat{X}|I} F \hat{X}[\Lambda(F, u)] \\ &= F \Lambda(F, u) \\ &= F (\text{Ad}_{F^{-1}}(U(0, g_0 \mathbf{e}_3) - N) + U(\omega, a) + N) \\ &= (U(0, g_0 \mathbf{e}_3) - N)F + F (U(\omega, a) + N) \end{aligned}$$

Be careful $\phi_F(\hat{X}) = \phi(\hat{X}, F) = F \hat{X} = \phi_{\hat{X}} F$.

Observer Formulation

The observer state kinematics are given by

$$\begin{aligned}\dot{\hat{X}} &= \hat{X} \Lambda(\hat{X}, u) \\ &= \hat{X} N - N \hat{X} + U(0, g_0 \mathbf{e}_3) \hat{X} + \hat{X} U(\omega_y, a_y)\end{aligned}$$

These are a copy of the system equation with measured inputs.

$$\begin{aligned}\omega_y &= \omega + \mu_\omega, & \mu_\omega &\sim \mathbf{N}(0, M_\omega) \\ a_y &= a + \mu_a, & \mu_a &\sim \mathbf{N}(0, M_a)\end{aligned}$$

We will assume that the kinematics N and input g_0 are known perfectly, although in certain situations adding additional noise into these parameters can be useful.

A key observation is that $\hat{X} \in \mathbf{SE}_2(3)$ is a group element, while $F \in \mathcal{SE}_2(3)$ lives on the torsor.

Error 1

The error

$$E = \phi(\hat{X}^{-1}, F) = F\hat{X}^{-1}$$

The error kinematics are

$$\begin{aligned}\dot{E} &= \dot{F}\hat{X}^{-1} - F\hat{X}^{-1}\dot{\hat{X}}\hat{X}^{-1} \\ &= F\Lambda(F, u)\hat{X}^{-1} - F\Lambda(\hat{X}, u)\hat{X}^{-1} \\ &= (U(0, g_0\mathbf{e}_3) - N)F\hat{X}^{-1} + F(U(\omega, a) + N)\hat{X}^{-1} \\ &\quad - F(U(\omega_y, a_y) + N)\hat{X}^{-1} - F\hat{X}^{-1}(U(0, g_0\mathbf{e}_3) - N)\end{aligned}$$

Note that

$$\begin{aligned}U(\omega_y, a_y) &= U(\omega, a) - U(\mu_\omega, \mu_a), \\ \mu &= (\mu_\omega, \mu_a, 0) \sim \mathbf{N}(0, \text{diag}(M_\omega, M_a, 0))\end{aligned}$$

Error

The error kinematics are

$$\begin{aligned}\dot{E} &= (U(0, g_0 \mathbf{e}_3) - N)F\hat{X}^{-1} - F\hat{X}^{-1}(U(0, g_0 \mathbf{e}_3) - N) \\
&\quad - FU(\mu_\omega, \mu_a)\hat{X}^{-1}, \quad \mu \sim \mathbf{N}(0, \text{diag}(M_\omega, M_a)) \\
&= U(0, g_0 \mathbf{e}_3)E - EU(0, g_0 \mathbf{e}_3) + EN - NE - E \text{Ad}_{\hat{X}} U(\mu_\omega, \mu_a), \\
&\quad \mu = (\mu_\omega, \mu_a, 0) \sim \mathbf{N}(0, \text{diag}(M_\omega, M_a, 0))\end{aligned}$$

The noise free kinematics are in “error system” form - that is they are independent of the trajectory of $\hat{X}$, F .

However, the noise term does depend on the observer trajectory $\hat{X}$

Linearisation

There are a number of ways to compute the linearisation of this equation. The simplest algebraically is to approximate

$$E \approx I + \epsilon^\wedge + \mathcal{O}(\epsilon^2)$$

where $\epsilon^\wedge \in \mathfrak{se}_2(3)$ and $\mu \sim \mathbf{N}(0, \text{diag}(M_\omega, M_a))$ are small.

The wedge operation $\wedge$ is a mapping from $\mathbb{R}^9$ into $\mathfrak{se}_2(3)$ used to correspond the matrix ODE elements to vector space elements.

$$\begin{aligned} \wedge : \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3 &\rightarrow \mathfrak{se}_2(3) & \vee : \mathfrak{se}_2(3) &\rightarrow \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3 \\ (\omega, a, v)^\wedge &:= \begin{pmatrix} \omega^\times & a & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} & \begin{pmatrix} \omega^\times & a & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}^\vee &= (\omega, a, v) \end{aligned}$$

Outline

- 1 Inertial Navigation Systems
 - Inertial Frames and Galilean Transformations
 - Galilean Kinematics
 - Gravity
 - Isochronous frames
 - INS kinematic model
- 2 Equivariant Filter
 - Problem Setup
 - Computing the Lift
 - Error formulation
- 3 Computing Error Linearisations
 - State Linearisation
 - Output Linearisation
- 4 Perspective

Linearisation

$$\dot{E} = U(0, g_0 \mathbf{e}_3)E - EU(0, g_0 \mathbf{e}_3) + EN - NE - E \operatorname{Ad}_{\hat{X}} \mu^\wedge$$

Since $\mu = (\mu_\omega, \mu_a, 0)$ and hence $U(\mu_\omega, \mu_a) = \mu^\wedge$. Then

$$\begin{aligned} \frac{d}{dt}(I + \epsilon) &= \dot{\epsilon}^\wedge \\ &= U(0, g_0 \mathbf{e}_3)(I + \epsilon^\wedge) - (I + \epsilon^\wedge)U(0, g_0 \mathbf{e}_3) \\ &\quad + (I + \epsilon^\wedge)N - N(I + \epsilon^\wedge) \\ &\quad - (1 + \epsilon) \operatorname{Ad}_{\hat{X}} \mu^\wedge + \mathbb{O}(\epsilon^2) \\ &\approx U(0, g_0 \mathbf{e}_3)\epsilon^\wedge - \epsilon^\wedge U(0, g_0 \mathbf{e}_3) + \epsilon^\wedge N - N\epsilon^\wedge - \operatorname{Ad}_{\hat{X}} \mu^\wedge \end{aligned}$$

Since $\epsilon^\wedge \operatorname{Ad}_{\hat{X}} \mu^\wedge = \mathbb{O}(\epsilon\mu)$.

Lie Bracket

Recall the Lie-bracket definition on $\mathfrak{gal}(3)$ For any two element $U, V \in \mathfrak{gal}(3)$ then

$$[U, V] = UV - VU = \text{ad}_U V$$

where these equations can be thought of as matrix multiplication.

Note that $N \notin \mathfrak{se}_2(3)$ although it is in $\mathfrak{gal}(3)$. However, we are saved by the fact that $\text{ad}_N U(\omega, a) \in \mathfrak{se}_2(3)$ - this is a particular property of the $N \in \mathfrak{gal}(3)$ matrix.

Linearisation

$$\dot{\epsilon}^\wedge = \text{ad}_{(U(0, g_0 e_3) + N)} \epsilon^\wedge - \text{Ad}_{\hat{X}} \mu^\wedge + \mathbb{O}(\epsilon^2, \epsilon \mu)$$

Linearisation

The linearisation

$$\dot{\epsilon}^\wedge = \text{ad}_{(U(0, g_0 \mathbf{e}_3) + N)} \epsilon^\wedge - \text{Ad}_{\hat{X}} \mu^\wedge$$

is written as a matrix ODE. We want to write this as a vector ODE

$$\dot{\epsilon} = A_t \epsilon + B_t \mu$$

That is, find linear matrices $A_t \in \mathbb{R}^{9 \times 9}$ and $B_t \in \mathbb{R}^{9 \times 9}$ such that

$$\begin{aligned} \text{ad}_{(U(0, g_0 \mathbf{e}_3) + N)} \epsilon^\wedge &= (A_t \epsilon)^\wedge \\ \text{Ad}_{\hat{X}} \mu^\wedge &= (B_t \mu)^\wedge \end{aligned}$$

We introduce the notation

$$A_t = \text{ad}_{(U(0, g_0 \mathbf{e}_3) + N)}^\vee, \quad B_t = \text{Ad}_{\hat{X}}^\vee$$

Exercise: Compute A_t and B_t .

Discretisation

The continuous time linearised error system is

$$\dot{\epsilon} = A_t \epsilon + B_t \mu$$

Then integrating this for zero noise

$$\epsilon_{k+1} = \exp(A_t dt) \epsilon_k$$

The he formulation as written does not really make sense for continuous noise! A better representation is to interpret μ as a Wiener process $d\mu$ and consider (details are not necessary)

$$d\epsilon = A_t \epsilon dt + B_t d\mu.$$

Integrating this one obtains

$$\epsilon_{k+1} = \exp(A_t dt) \epsilon_k + B_t \mu_k, \quad \mu_k \sim \mathbf{N}(0, dtM)$$

Output

The output that we consider for the INS problem is position

$$y = p + \mu_y, \quad \mu_y \sim \mathbf{N}(0, N_y)$$

Recall that the state $F = (R, v, p) \in \mathcal{SE}_2(3)$ and then

$$\bar{p} = \begin{pmatrix} p \\ 0 \\ 1 \end{pmatrix} = h(F) = F \mathbf{e}_5 = \begin{pmatrix} R & v & p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{e}_5$$

It is possible to extract the actual state p from $\bar{p}$ but this would 'hide' the state F inside the h function. As written, the state F acts linearly on a reference $\mathbf{e}_5$ to generate $\bar{p}$.

Spatial Equivariance of output

The output $\bar{y}$ is a **spatial** output

$$\bar{p} = F\mathbf{e}_5$$

Consider a left group action $\alpha : \mathbf{SE}_2(3) \times \mathcal{SE}_2(3) \rightarrow \mathcal{SE}_2(3)$ defined by

$$\alpha(\hat{X}, F) = \hat{X}F$$

then

$$h(\alpha(\hat{X}, F)) = \hat{X}F\mathbf{e}_5 = \hat{X}h(F)$$

and one can define a left group action $\rho^L : \mathbf{SE}_2(3) \times \overline{\mathbb{R}}^3 \rightarrow \overline{\mathbb{R}}^3$ on the output space $\mathcal{N} = \overline{\mathbb{R}}^3$ by

$$\rho^L(\hat{X}, \bar{y}) = \hat{X}\bar{y}$$

to obtain left equivariance

$$\rho^L(\hat{X}, h(F)) = h(\alpha(\hat{X}, F))$$

Body output

The equivariant filter framework requires that the state and output symmetries have the same handedness!

We can generate a right-equivariant output from a left-equivariant output by working with a **pseudo output** $\mathring{y}$

$$\mathring{y} = \mathbf{e}_5 = F^{-1}y = h_y(F)$$

Here

- $\mathring{y}$ is used as the output measurement at each update.
- The output function h_y is a parameterised family of outputs that depend on the actual measured output.

This process formally introduces stochastic dependence in the noise model for $\mathring{y}$, however, this dependence can be managed in the algebra of the linearisation.

Equivariance of the pseudo output

Let $\mathring{\bar{y}} = h_y(F) = F^{-1}\bar{y}$

Define a (right-handed) output group action

$\rho : \mathbf{SE}_2(3) \times \overline{\mathbb{R}}^3 \rightarrow \overline{\mathbb{R}}^3$ on the output space $\mathcal{N} = \overline{\mathbb{R}}^3$ by

$$\rho(\hat{X}, \mathring{\bar{y}}) = \hat{X}^{-1}\bar{y}$$

Then

$$h_y(\phi(\hat{X}, F)) = h_y(F\hat{X}) = \hat{X}^{-1}F^{-1}y = \rho(\hat{X}, h_y(F))$$

That is, the pseudo output is equivariant.

Output Error

Define the **output error**

$$\bar{d} = \rho(\hat{X}^{-1}, h_y(F)) = \rho(\hat{X}, \overset{\circ}{y})$$

A key observation is that d is computable!

Then one has

$$\bar{d} = \rho(\hat{X}^{-1}, h_y(F)) = h_y(\phi_{\hat{X}^{-1}}(F)) = h_y(F\hat{X}^{-1}) = \hat{X}F^{-1}\bar{y}$$

Recall that $E = \phi_{\hat{X}^{-1}}(F)$ then

$$\bar{d} = E^{-1}\bar{y}$$

Equivariant innovation

Recall that $\bar{d} = E^{-1}\bar{y}$ then substitute $E^{-1} \approx I_5 - \epsilon^\wedge$ to get

$$\begin{aligned}\bar{d} &= (I_5 - \epsilon^\wedge)\bar{y} + \mathbb{O}(\epsilon^2) \\ &= \bar{y} - \epsilon^\wedge\bar{y} + \mathbb{O}(\epsilon^2)\end{aligned}$$

The equivariant innovation is defined as

$$\begin{aligned}\tilde{d} &= \bar{d} - \bar{p} \\ &= \bar{y} - \epsilon^\wedge\bar{y} - \bar{p} + \mathbb{O}(\epsilon^2) \\ &= -\epsilon^\wedge\bar{y} + \mu_y + \mathbb{O}(\epsilon^2, \epsilon\mu_y)\end{aligned}$$

where here we have used $\bar{y} = \bar{p} + \mu_y$ where $\mu_y \sim \mathbf{N}(0, N)$

Equivariant innovation model

Then writing this as an innovation model as a linear function of error variable one has

$$\tilde{d} = -\epsilon^{\wedge} \bar{y} + \mu_y, \quad \mu_y \sim \mathbf{N}(0, N_y)$$

The linearisation is defined implicitly by finding $C_t \in \mathbb{R}^{5 \times 9}$ such that

$$\tilde{d} = C_t \epsilon = -\epsilon^{\wedge} \bar{y}$$

This is simple a case of re-ordering columns and rows. **Excercise!**

Outline

- 1 Inertial Navigation Systems
 - Inertial Frames and Galilean Transformations
 - Galilean Kinematics
 - Gravity
 - Isochronous frames
 - INS kinematic model
- 2 Equivariant Filter
 - Problem Setup
 - Computing the Lift
 - Error formulation
- 3 Computing Error Linearisations
 - State Linearisation
 - Output Linearisation
- 4 Perspective

Summary

From this construction we have computed

$$\begin{aligned}\epsilon_{k+1} &= \mathbf{A}_k \epsilon_k + \mathbf{B}_k \mu_k, & \mu_k &\sim \mathbf{N}(0, M_k) \\ \tilde{d}_{k+1} &= \mathbf{C}_t \epsilon_{k+1} + \mu_y, & \mu_y &\sim \mathbf{N}(0, N)\end{aligned}$$

Where

$$\mathbf{A}_k = \exp \left(\text{ad}_{(U(0, g_0 \mathbf{e}_3) + N)}^\vee dt \right), \quad \mathbf{B}_k = \text{Ad}_{\hat{X}_k}^\vee,$$

$$\mathbf{C}_k \epsilon_k = -\epsilon_k^\wedge \bar{y}_k$$

From this - you can run a Kalman filter in parallel with the observer update $\hat{X}_k$

Equivariant Filter

Recall the form for the Equivariant Filter

Predict:

$$\begin{aligned}\hat{X} &\leftarrow \hat{X} \exp \left(\Lambda(\phi_{\hat{\xi}}(\hat{X}), u) dt \right), \\ \Sigma &\leftarrow \mathbf{A}_t \Sigma \mathbf{A}_t^\top + B_t \mathbf{M} B_t^\top,\end{aligned}$$

Update:

$$\begin{aligned}K &:= \left(\Sigma \mathbf{C}^\top (\mathbf{C} \Sigma \mathbf{C}^\top + \mathbf{N})^{-1} \right) \\ \Sigma &\leftarrow (I - K \mathbf{C}) \Sigma, \\ \Delta &:= D_{E|I} \phi_{\hat{\xi}}(E)^\dagger K \rho(\hat{X}^{-1}, y), \\ \hat{X} &\leftarrow \exp(\Delta) \hat{X},\end{aligned}$$

General form

What have we actually computed.

The general form of the EqF is written as a set of differentials that encode the linearisation computations

$$\begin{aligned}\dot{\varepsilon} &= A_t^0 \varepsilon, \\ A_t^0 &= D_{e|\xi} \vartheta(e) \cdot D_{E|I} \phi_\xi(E) \cdot D_{e|\xi} \Lambda(e, \dot{u}) \cdot D_{\varepsilon|0} \vartheta^{-1}(\varepsilon), \\ \delta(h(e)) &= \delta\left(h\left(\rho(\hat{X}^{-1}, y)\right)\right) \approx C^0 \varepsilon, \\ C^0 &= D_{y|\hat{y}} \delta(y) D_{e|\xi} h(e) D_{\varepsilon|0} \vartheta^{-1}(\varepsilon).\end{aligned}$$

However, since we had nice algebraic expressions for the various error variables, we were able to compute the linearisations explicitly (and nicely).

What about the projection

What happens to the pseudo inverse $D_{E|I} \phi_{\xi}^{\dagger}$.

Recall that $\phi_{\dot{F}}(\hat{X}) = \dot{F}\hat{X} = \hat{X}$ since $\dot{F} = I_5$.

So $\phi_{\dot{F}} = \text{Identity}$ and its inverse is identity as well. So this disappeared in the analysis.